CMPG325

TECHNICAL REPORT

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1. Introduction

This report explains the implementation and testing of the network designed for Dijo Tsa Rona Foods Processing. The network was designed using Cisco Packet Tracer. The main purpose of the network is to connect the different areas of the organisation while keeping the departments separated using VLANs. The network includes the production network, office network, server network, management network and public Wi-Fi network.

The network uses the 192.168.25.0/24 address block. VLSM was used to divide the address block into different size networks according to the requirements. The client also requested a second internet connection. ISP1 is used as the main internet connection and ISP2 is used as the backup connection. The implementation was tested using ping tests, routing checks, HSRP checks and VLAN checks.

2. Client Requirements

The network must:

- Connect the office users.
- Connect the production area.
- Provide access to servers.
- Provide a management network for network devices.
- Provide wireless access in the allowed public areas.
- Allow communication between the required networks.
- Use the 192.168.25.0/24 address block.
- Use VLSM to avoid wasting IP addresses.
- Have two internet connections.
- Be designed and tested in Cisco Packet Tracer.
- Continue using the second internet connection if the main internet connection fails.

The wireless network is only required in public areas and is not intended to cover the whole factory. The second internet connection was added because of the change request CR15. The purpose of the second connection is to provide another connection if the main internet connection fails.

3. Network Implementation

The network was implemented using two main routers, switches, end devices, a wireless access point and two simulated internet service provider routers.

The main routers are:

- R1-Main
- R2-Backup

The internet connections are:

- ISP1
- ISP2

R1-Main is connected to ISP1 and R2-Backup is connected to ISP2.
R1 and R2 are also connected to each other using a routed link.
The switches are used to connect the different departments and network devices.
The network was divided into VLANs so that the different areas have their own network.

4. VLAN Configuration

The following VLANs were implemented:

VLAN	Name	Network	Gateway
10	PRODUCTION	192.168.25.0/26	192.168.25.1
20	OFFICE	192.168.25.64/27	192.168.25.65
30	SERVER	192.168.25.96/28	192.168.25.97
40	PUBLIC-WIFI	192.168.25.112/28	192.168.25.113
50	MANAGEMENT	192.168.25.128/29	192.168.25.129

These networks follow the VLSM addressing plan from Milestone 1.
VLAN 10 is used for the production area.
VLAN 20 is used for office users.
VLAN 30 is used for servers.
VLAN 40 is used for the public wireless network.
VLAN 50 is used for switch management. The switches have management IP addresses assigned to their VLAN 50 interfaces. A router gateway for this VLAN is not currently configured.

5. IP Addressing

R1-Main

Interface	IP Address	Purpose
G0/0.10	192.168.25.2	Production
G0/0.20	192.168.25.66	Office
G0/0.30	192.168.25.98	Server
G0/0.40	192.168.25.114	Public Wi-Fi
G0/1	192.168.25.137	Router-to-router
G0/2	203.0.113.1	ISP1

R2-Backup

Interface	IP Address	Purpose
G0/0.10	192.168.25.3	Production
G0/0.20	192.168.25.67	Office
G0/0.30	192.168.25.99	Server
G0/0.40	192.168.25.115	Public Wi-Fi
G0/1	192.168.25.138	Router-to-router
G0/2	198.51.100.1	ISP2

The router-to-router network is 192.168.25.136/30.

6. Switch Management

The switches use VLAN 50 for management.

The current switch management addresses are:

Switch	Management IP
SW-Core	192.168.25.130
SW-Office	192.168.25.131
SW-Production	192.168.25.132
SW-Server	192.168.25.133

The management VLAN allows the network devices to have their own management network.

7. HSRP Implementation

HSRP was configured between R1-main and R2-Backup to provide gateway redundancy.
R1-Main has a higher HSRP priority of 110 and is the active router.
R2-Backup has a priority of 100 and acts as the standby router.

The virtual gateway addresses are:

VLAN	Virtual Gateway	R1	R2
10	192.168.25.1	192.168.25.2	192.168.25.3
20	192.168.25.65	192.168.25.66	192.168.25.67
30	192.168.25.97	192.168.25.98	192.168.25.99
40	192.168.25.113	192.168.25.114	192.168.25.115

HSRP was configured on VLAN 10, 20, 30 and 40. The **show standby brief** command was used to confirm that R1 is active and R2 is standby. The HSRP configuration provides a virtual default gateway for the users instead of requiring users to use the physical router address.

8. Public Wi-Fi

A separate Public Wi-Fi network was implemented using VLAN 40.

The Public Wi-Fi network is:

Network: 192.168.25.112/28
Gateway: 192.168.25.113

The guest laptop was given:

IP address: 192.168.25.120
Subnet mask: 255.255.255.240
Default Gateway: 192.168.25.113

The wireless access point is connected to the switch on an access port configured for VLAN 40. The guest laptop connects to the access point wirelessly.

9. Routing

Static routing was used to allow communication between the internal network and the second internet connection.

R1-Main has a route to ISP2 network through R2:

ip route 198.51.100.0 255.255.255.252 192.168.25.138

R2-Backup uses ISP2 as its default route:

ip route 0.0.0.0 0.0.0.0 198.51.100.2

A backup default route was also configured for R1:

ip route 0.0.0.0 0.0.0.0 192.168.25.138 10

The administrative distance of 10 makes this a floating static route. The normal ISP1 route is preferred while it is available.

ISP2 also has a route back to the internal network:

ip route 192.168.25.0 255.255.255.0 198.51.100.1

This allows devices behind ISP2 to reach the internal network.

10. Physical Connections

The physical topology was implemented on Cisco Packet Tracer.

The important connections include:

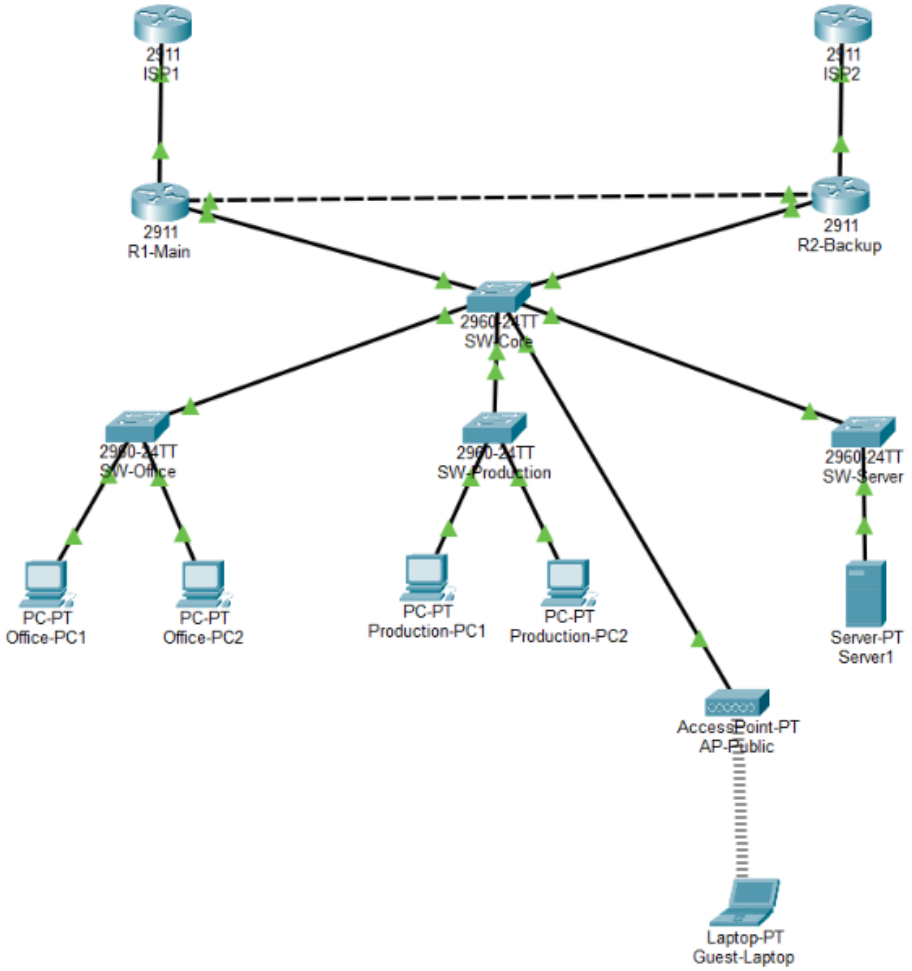
Device	Interface	Connected to	Interface
R1-Main	G0/1	R2-Backup	G0/1
R1-Main	G0/2	ISP1	G0/0
R2-Backup	G0/2	ISP2	G0/0
Guest AP	Ethernet port	SW-Core	Fa0/4
Guest Laptop	Wireless	Guest AP	Wireless

The switch trunk ports were checked using: **show interfaces trunk**

SW-Core showed the trunk ports as active and VLANs 10, 20, 30, 40 and 50 were allowed.
For the Ethernet connections, copper straight-through cables were used for normal router-to-switch, switch-to-device and access point connections. Direct router-to-

router Ethernet links can use crossover cables where required by the Packet Tracer design.

11. Physical Topology



12. Testing

Test was performed to check that the network was working correctly.

The tests included:

- Router connectivity
- HSRP
- VLAN gateways
- Guest Wi-Fi
- ISP connectivity
- Routing
- Return routing
- Failover

12.1 R1 to ISP1

The first test was from R1-Main to ISP1

```
R1-Main#ping 203.0.113.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 203.0.113.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

R1-Main#
```

The ping was successful with 100% replies.

12.2 R1 to R2

The connection between the two routers was tested.

```
R1-Main#ping 192.168.25.138

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.25.138, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

R1-Main#
```

The ping was successful.

12.3 HSRP Gateway testing

The HSRP virtual gateways were tested from R1.

```
R1-Main#ping 192.168.25.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.25.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/22/68 ms

R1-Main#ping 192.168.25.65

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.25.65, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/29/103 ms

R1-Main#ping 192.168.25.97

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.25.97, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/11/30 ms

R1-Main#ping 192.168.25.113

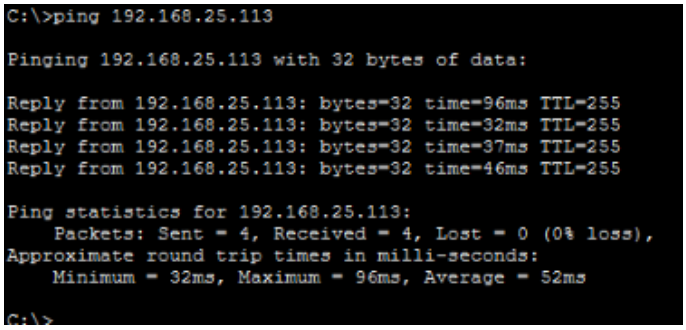
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.25.113, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/18/56 ms

R1-Main#
```

All tested gateways were reachable.

12.4 Guest Laptop to Gateway

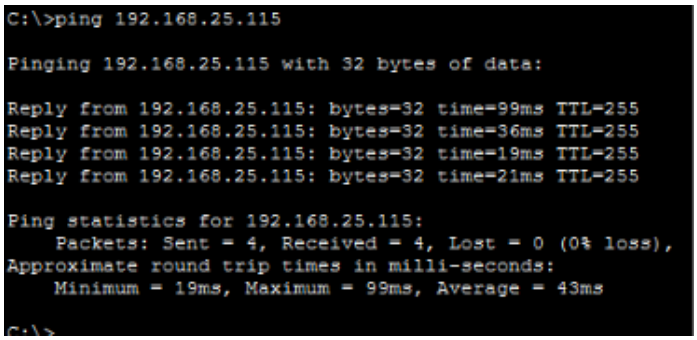
The guest laptop was tested against its gateway



The result was 4 replies with 0% packet loss.

12.5 Guest Laptop to R2

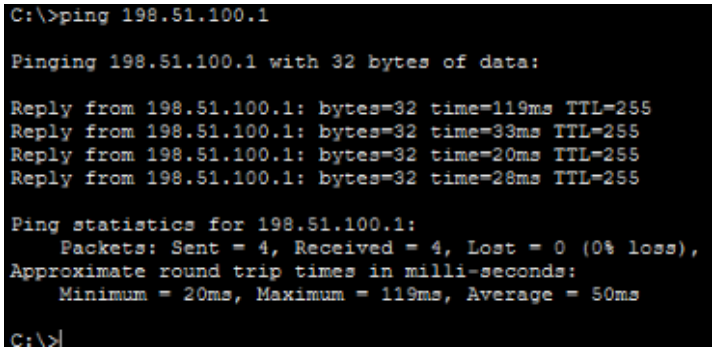
The guest laptop was tested against the R2 VLAN 40 interface



The result was 4 replies with 0% packet loss.

12.6 Guest Laptop to ISP2

The guest laptop was tested against the R2 side of the second internet connection.



The result was successful with 4 replies and 0% packet loss.

12.7 ISP2 to Guest Laptop

A return traffic test was performed from ISP2 to the guest laptop.

```
ISP2>enable
ISP2#ping 192.168.25.120

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.25.120, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 20/28/33 ms

ISP2#
```

The result was successful with 100% replies.

12.8 HSRP Failover Testing

HSRP failover was tested by temporarily disabling the VLAN 40 interface on R1.

The following configuration was used for testing:

```
R1-Main#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1-Main(config)#interface gigabitEthernet0/0.40
R1-Main(config-subif)#shutdown

R1-Main(config-subif)#
 %HSRP-6-STATECHANGE: GigabitEthernet0/0.40 Grp 40 state Active -> Init

%LINK-5-CHANGED: Interface GigabitEthernet0/0.40, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.40, changed state to down

R1-Main(config-subif)#end
R1-Main#
%SYS-5-CONFIG_I: Configured from console by console

R1-Main#
```

The HSRP status was then checked:

The result showed that R2-Backup became Active for VLAN 40 after the R1 interface was disabled. The HSRP virtual gateway remained reachable during the failover test.

```
R2-Backup>enable
R2-Backup#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State    Active          Standby         Virtual IP
Gig         10   100 P Standby  192.168.25.2    local           192.168.25.1
Gig         20   100 P Standby  192.168.25.67   local           192.168.25.65
Gig         30   100 P Standby  192.168.25.98   local           192.168.25.97
Gig         40   100 P Active   local           192.168.25.115  192.168.25.113
R2-Backup#
```

The guest laptop was then tested against the HSRP virtual gateway:

```
C:\>ping 192.168.25.113

Pinging 192.168.25.113 with 32 bytes of data:

Reply from 192.168.25.113: bytes=32 time=146ms TTL=255
Reply from 192.168.25.113: bytes=32 time=38ms TTL=255
Reply from 192.168.25.113: bytes=32 time=7ms TTL=255
Reply from 192.168.25.113: bytes=32 time=47ms TTL=255

Ping statistics for 192.168.25.113:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 7ms, Maximum = 146ms, Average = 59ms

C:\>
```

After the test, the R1 interface was restored:

```
R1-Main#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1-Main(config)#interface gigabitEthernet0/0.40
R1-Main(config-subif)#no shutdown

R1-Main(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.40, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.40, changed state to up

R1-Main(config-subif)#en
 %HSRP-6-STATECHANGE: GigabitEthernet0/0.40 Grp 40 state Standby -> Active
d
R1-Main#
%SYS-5-CONFIG_I: Configured from console by console

R1-Main#
```

The HSRP status was checked again:

```
R2-Backup#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State    Active          Standby         Virtual IP
Gig         10   100 P Standby  192.168.25.2    local           192.168.25.1
Gig         20   100 P Standby  192.168.25.67   local           192.168.25.65
Gig         30   100 P Standby  192.168.25.98   local           192.168.25.97
Gig         40   100 P Standby  192.168.25.114  local           192.168.25.113
R2-Backup#
```

13. Internet Failover Testing

The network was designed with a primary and backup internet connection.
The normal path is through ISP1.
The backup path is through R2 and ISP2.
R1 is a floating static default route through: ip route 0.0.0.0 0.0.0.0 192.168.25.138 10

The internet failover was tested by temporarily shutting down R1's ISP interface.

The interface used for the test was:

```
R1-Main#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1-Main(config)#interface gigabitEthernet0/2
R1-Main(config-if)#shutdown

R1-Main(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down

R1-Main(config-if)#end
R1-Main#
%SYS-5-CONFIG_I: Configured from console by console

R1-Main#
```

The routing table was then checked:

```
R1-Main#show ip route 0.0.0.0
Routing entry for 0.0.0.0/0, supernet
Known via "static", distance 10, metric 0, candidate default path
  Routing Descriptor Blocks:
  * 192.168.25.138
      Route metric is 0, traffic share count is 1

R1-Main#
```

The routing table showed that the backup default route through 192.168.25.138 became active after the ISP1 interface was disabled.

The connection to ISP2 was then tested:

```
R1-Main#ping 198.51.100.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 198.51.100.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms
```

After the test, the ISP1 interface was restored:

```
R1-Main#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1-Main(config)#interface gigabitEthernet0/2
R1-Main(config-if)#no shutdown

R1-Main(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

R1-Main(config-if)#end
R1-Main#
%SYS-5-CONFIG_I: Configured from console by console
```

The routing table was checked again:

```
R1-Main#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 203.0.113.2 to network 0.0.0.0

     192.168.25.0/24 is variably subnetted, 10 subnets, 5 masks
C       192.168.25.0/26 is directly connected, GigabitEthernet0/0.10
L       192.168.25.2/32 is directly connected, GigabitEthernet0/0.10
C       192.168.25.64/27 is directly connected, GigabitEthernet0/0.20
L       192.168.25.66/32 is directly connected, GigabitEthernet0/0.20
C       192.168.25.96/28 is directly connected, GigabitEthernet0/0.30
L       192.168.25.98/32 is directly connected, GigabitEthernet0/0.30
C       192.168.25.112/28 is directly connected, GigabitEthernet0/0.40
L       192.168.25.114/32 is directly connected, GigabitEthernet0/0.40
C       192.168.25.136/30 is directly connected, GigabitEthernet0/1
L       192.168.25.137/32 is directly connected, GigabitEthernet0/1
     198.51.100.0/30 is subnetted, 1 subnets
S       198.51.100.0/30 [1/0] via 192.168.25.138
     203.0.113.0/24 is variably subnetted, 2 subnets, 2 masks
C       203.0.113.0/30 is directly connected, GigabitEthernet0/2
L       203.0.113.1/32 is directly connected, GigabitEthernet0/2
S*   0.0.0.0/0 [1/0] via 203.0.113.2

R1-Main#
```

VLAN Configuration

SW-Core:

```
SW-Core#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
10   PRODUCTION                       active
20   OFFICE                           active
30   SERVER                           active
40   PUBLIC-WIFI                      active    Fa0/4
50   MANAGEMENT                       active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SW-Core#
```

Trunk Configuration

SW-Core:

```
SW-Core#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1
Fa0/2       on           802.1q         trunking      1
Fa0/3       on           802.1q         trunking      1
Gig0/1      on           802.1q         trunking      1
Gig0/2      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       1-1005
Fa0/2       1-1005
Fa0/3       1-1005
Gig0/1      1-1005
Gig0/2      1-1005

Port        Vlans allowed and active in management domain
Fa0/1       1,10,20,30,40,50
Fa0/2       1,10,20,30,40,50
Fa0/3       1,10,20,30,40,50
Gig0/1      1,10,20,30,40,50
Gig0/2      1,10,20,30,40,50

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       1,10,20,30,40,50
Fa0/2       1,10,20,30,40,50
Fa0/3       1,10,20,30,40,50
Gig0/1      1,10,20,30,40,50
Gig0/2      1,10,20,30,40,50

SW-Core#
```

Guest Laptop

```
C:\>ipconfig

Bluetooth Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.....: ::
   IPv6 Address.....: ::
   IPv4 Address.....: 0.0.0.0
   Subnet Mask.....: 0.0.0.0
   Default Gateway.....: ::
                                 0.0.0.0

Wireless0 Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.....: FE80::290:CFF:FED9:C409
   IPv6 Address.....: ::
   IPv4 Address.....: 192.168.25.120
   Subnet Mask.....: 255.255.255.240
   Default Gateway.....: ::
                                 192.168.25.113

C:\>
```

14. Conclusion

The network from Milestone 1 was implemented in Cisco Packet Tracer. The implementation includes separate VLANs for production, office users, servers, public Wi-Fi and network management. The HSRP was configured between the two main routers to provide gateway redundancy for the main user VLANs. A separate public Wi-Fi network was implemented using VLAN 40.

Two internet connections were also implemented. ISP1 is the main connection and ISP2 provides the backup connection. Testing was performed using ping tests, routing commands and HSRP verification. The failover tests showed that the network can use the backup connection when the main internet connection is unavailable.

15. Github Repository Link

https://github.com/lizziebusisiwe0410/CMPG325-Network-Design-Project